



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 5 Report

Internship Program — Batch 2026

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Submitted To:

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1. M-Tech Production & Marketing Internship Program 2026

Daily Internship Report

Date: 17 July 2026

2. 1. Introduction

This report summarizes the activities completed during the **M-Tech Production & Marketing AI/ML Internship Program** on **17 July 2026**. During today's session, I successfully studied and completed the **Neural Networks** experiment from the **Artificial Intelligence & Machine Learning Lab Manual**. The experiment introduced the fundamental concepts of **Artificial Neural Networks (ANNs)**, which form the foundation of **Deep Learning**. I learned how interconnected artificial neurons process information, recognize patterns, and continuously improve their predictions through training, enabling modern Artificial Intelligence systems to solve complex real-world problems.

3. 2. Objectives

The primary objectives of today's learning session were:

- To understand the fundamentals of **Artificial Neural Networks (ANNs)**.
 - To study the architecture of **input, hidden, and output layers**.
 - To learn the concepts of **forward propagation** and **backpropagation**.
 - To understand the roles of **weights, biases, and activation functions**.
 - To explore optimization techniques used in neural network training.
 - To examine practical applications of Neural Networks in Artificial Intelligence.
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4. 3. Task Description

Today's task involved studying the **Neural Networks** experiment from the AI/ML Lab Manual. The experiment focused on understanding the internal architecture and learning process of Artificial Neural Networks.

I studied how input data passes through multiple network layers, how neurons perform mathematical computations using weights and biases, and how predictions are generated through forward propagation. The experiment also explained how prediction errors are minimized using **backpropagation**, allowing the network

to continuously improve its performance during training. Additionally, I explored commonly used activation functions and learned how they help neural networks solve complex non-linear problems.

5. 4. Work Done

During today's internship session, I successfully completed the **Neural Networks** experiment from the AI/ML Lab Manual.

I began by studying the architecture of **Artificial Neural Networks** and understanding how multiple interconnected neurons work together to process input data. I learned the purpose of the **input layer**, **hidden layers**, and **output layer**, and understood how information flows through each stage of the network.

Next, I explored the importance of **weights** and **biases** in determining how input features influence the network's predictions. I learned how these parameters are adjusted during training to improve the model's accuracy.

I then studied **forward propagation**, where input data moves through the neural network to generate predictions, followed by **backpropagation**, where prediction errors are propagated backward through the network to update weights and minimize future errors. This helped me understand the learning mechanism that enables neural networks to improve over time.

I also explored several widely used **activation functions**, including **ReLU (Rectified Linear Unit)**, **Sigmoid**, and **Softmax**, and learned how they introduce non-linearity into the model, enabling neural networks to solve complex prediction and classification problems.

Finally, I examined several real-world applications of Artificial Neural Networks, including **image recognition**, **speech recognition**, **natural language processing**, **recommendation systems**, **robotics**, **healthcare**, and intelligent decision-making systems. These examples demonstrated the versatility and importance of neural networks in modern Artificial Intelligence.

6. 5. Challenges Faced

The concepts presented in the **Neural Networks** experiment were highly informative and formed an essential part of Deep Learning. However, understanding how **backpropagation** updates network parameters through gradients required additional study and careful analysis.

Learning the mathematical relationship between **weights**, **biases**, and **gradient calculations** also required extra attention. In addition, understanding when and why different **activation functions** should be used initially posed some difficulty. By reviewing theoretical concepts, studying practical examples, and analyzing the complete neural network training process, I developed a much clearer understanding of these topics.

7. 6. Key Learnings

During today's learning session, I achieved the following outcomes:

- Developed a strong understanding of the fundamentals of **Artificial Neural Networks (ANNs)**.
- Learned the architecture and working principles of **input**, **hidden**, and **output** layers.
- Understood how **forward propagation** generates predictions and how **backpropagation** minimizes prediction errors.
- Gained knowledge of the roles of **weights**, **biases**, and **activation functions** in neural network learning.
- Learned the importance of activation functions such as **ReLU**, **Sigmoid**, and **Softmax**.
- Explored practical applications of Neural Networks in **computer vision**, **speech recognition**, **natural language processing**, **robotics**, **healthcare**, and **recommendation systems**.
- Strengthened my understanding of the core concepts that serve as the foundation of modern Deep Learning.

8. 7. Conclusion

The successful completion of the **Neural Networks** experiment provided valuable theoretical knowledge and practical insight into one of the most important areas of Artificial Intelligence and Deep Learning. I developed a comprehensive understanding of how Artificial Neural Networks process information, learn from data, and continuously improve their predictions through training.

The knowledge and practical skills gained during today's session have significantly strengthened my understanding of Deep Learning and prepared me for studying more advanced neural network architectures and AI models. This experiment has established a strong foundation for future work in Artificial Intelligence, Machine Learning, and intelligent data-driven systems throughout the internship program.

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13. Intern Information

Field	Details
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Date	17 July, 2026

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